

CHAPTER ONE

INTRODUCTION

1.1 Background of Study

The widespread integration of information technology into everyday life has led to an increased need for processing vehicle data within information systems. Since an information system is of little use without relevant data, it becomes essential to translate vehicle information from the physical world into these systems. This can be done either by human operators or by intelligent systems that can recognize vehicles through their license plates and feed this data into the system. As a result,

many recognition techniques have emerged, and license plate recognition systems are now widely used in various traffic and security contexts, such as parking management, access control, border security, and tracking stolen vehicles (Li et al.,

2019).

A vehicle registration plate, usually made of metal or plastic, is attached to a vehicle for official identification purposes. The registration number, which may consist of numbers or letters, is used to uniquely identify the vehicle within a regional or national database. Depending on the issuing authority, the identifier may either be unique across an entire country or limited to a specific state or region. In most places, plates must be displayed on both the front and rear of the vehicle, although in some cases, like for motorboats, only one plate is required (Fang et al., 2016). These identifiers are linked to detailed information about the vehicle, including its make, model, color, year of manufacture, engine specifications, and the registered owner's details (Smith & Smith, 2021).

Access control refers to restricting entry to authorized individuals, which can be achieved through mechanical means (e.g., locks and keys) or technological systems. Automatic Car Plate Recognition (ACPR) is a mass surveillance technology that uses Optical Character Recognition (OCR) to read vehicle registration plates from images. ACPR systems can utilize existing closed-circuit television (CCTV) or specialized cameras for the task. These systems are employed by law enforcement and in electronic toll collection on pay-per-use roads, as well as for monitoring traffic and individuals (Zhao et al., 2020).

ACPR systems typically store captured images and license plate text, and some are capable of capturing photographs of drivers. Infrared lighting allows these systems to function effectively at any time of day. ACPR technology is often region-specific due to variations in license plate designs (Dai & Liu, 2018).

The software component of ACPR systems runs on standard computer hardware and can interface with other applications or databases. The system first uses image manipulation techniques to detect, normalize, and enhance the license plate image, followed by OCR to extract the alphanumeric characters. ACPR systems can operate in two primary modes: real-time processing at the lane location or batch processing where images are transmitted to a remote server for later analysis. Real-time processing typically completes in about 250 milliseconds, with data either stored locally or transmitted to a remote server. In contrast, batch processing involves high bandwidth requirements for transmitting images to servers (Yang et al., 2021).

During the 1990s, advancements in technology transformed ACPR systems from expensive, fixed installations to more affordable, mobile solutions. This evolution was driven by the development of software that could run on non-specialist, cost-effective PC hardware, eliminating the need for pre-defined camera angles, directions, sizes, and speeds (Chen et al., 2017).

1.2 Statement of the Problem

Over the years, vehicles have been involved in various criminal activities, including robbery, kidnapping, and auto theft, which continue to be significant issues for law enforcement agencies (Kang et al., 2022). The adoption of Automated Car Plate Recognition (ACPR) systems has become increasingly prevalent as a solution to these challenges. ACPR systems automatically capture images of vehicle license plates, convert these images into alphanumeric characters using Optical Character Recognition (OCR) or similar technology, and compare the extracted plate numbers against databases of interest. This process allows for rapid identification and alerting of law enforcement officers when a vehicle of interest is detected (Huang & Li, 2021).

Despite the automated nature of ACPR systems, human intervention remains crucial. Officers must verify the accuracy of the license plate reading, ensure that the plate is issued from the appropriate jurisdiction, and confirm the validity of any alerts. The need for technological advancements persists to enhance enforcement and investigative capabilities, improve data collection, and streamline the comparison of vehicle license plates with databases of stolen or wanted vehicles. Law enforcement personnel continually seek to identify vehicles associated with criminal activities, outstanding violations, or other legal issues, and ACPR systems play a vital role in facilitating these efforts (Kang et al., 2022; Huang & Li, 2021).

1.3 Aim and Objectives of the study

1.3.1 Aim

The aim of this research work is to develop an application that will provide access control through automated car plate recognition.

1.3.2 Objectives

The objectives of this study are as follows;

1. To assist law enforcement agencies in tracking stolen vehicles
2. To design a database that will serve as a repository for car plate numbers already registered.
3. To design a responsive system for crime detection
4. To ensure quick collection of data about vehicles within the scope of study.

1.4 Significance of the study

The proposed system will be of great advantage to Rivers state law enforcement agency, Rivers State Internal Revenue Service and other relevant authorities in the following ways;

- ACPR can assist the police to identify persons of interest associated with criminal activity. According to a recent article, auto theft is associated with a wide range of criminal activities, including the offences of break and enter, armed robbery, and drug-related offences. In other words, offenders steal cars for use in the commission of additional offences. For example, drug offenders may steal from within cars, but may also steal the car to raise money to purchase drugs.
- Improved Performance and Efficiency: There is no comparison between the number of plates an officer on patrol entering license plate information manually into an onboard computer and the number that automated ACPR technology can handle. With the potential ability to read up to 3,000 plates an

hour, this technology can improve productivity. Not only does the system read plates rapidly, but hits are collected only on vehicles displaying license plates that match the desired criteria and appear in the database.

- **Increased Crime Detection:** The substantially larger number of vehicles pulled over as a result of ACPR technology in patrol cars means that officers will come face to face with more criminals, generating more arrests. Officers attached to ACPR teams make 10 times more arrests than non-ACPR members. Arrests that are the result of ACPR stops are primarily for vehicle crimes, robbery, theft, burglary, and drug offenses. It is well known that car thieves steal cars not just to go for a drive but to help them commit other crimes such as breaking and entering, robbery, home invasion, and drug trafficking. In this way, ACPR technology can be valuable in preventing or solving many types of crime plaguing society.

1.5 Scope of the study

Every project is carried out to achieve a set of goals with some conditions keeping in mind that it should be easy to use, feasible and user friendly. As the goal of this project is to develop an application that will provide access control through automated car plate recognition, this system will be designed keeping in mind the conditions (easy to use, feasibility and user friendly) stated above. The proposed project's intended users are; Rivers State Law enforcement agencies, Rivers State Internal Revenue Service and other relevant agencies.

1.6 Definition of terms

1. **Optical character recognition (OCR):** This is the mechanical or electronic conversion of images of typewritten or printed text into machine-encoded text. It is widely used as a form of data entry from printed paper data records. It is a common method of digitizing printed texts so that it can be electronically edited, searched, stored more compactly, displayed on-line, and used in machine processes such as machine translation, text-to-speech, key data and text mining. OCR is a field of research in pattern recognition, artificial intelligence and computer vision.

2. **Automatic Car Plate Recognition (ACPR):** This is a mass surveillance method that uses optical character recognition on images to read vehicle registration plates. They can use existing closed-circuit television or road-rule enforcement cameras or ones specifically designed for the task.

3. **Surveillance:** This is the monitoring of the behavior, activities, or other changing information, usually of people for the purpose of influencing, managing, directing, or protecting them. This can include observation from a distance by means of electronic equipment (such as CCTV cameras) or interception of electronically transmitted information using no, low or high technology methods.

4. **Car Plate:** A vehicle registration plate is a metal or plastic plate attached to a motor vehicles for official identification purposes. The registration identifier is a numeric or alphanumeric code that uniquely identifies the vehicle within the issuing region's database.

5. **Automation:** This is the use of various control systems for operating equipment such as machinery, processes in factories, boilers and heat-treating ovens, switching in telephone networks, steering and stabilization of ships, aircraft and other applications with minimal or reduced human intervention. Most processes today have been completely automated. The biggest benefit of automation is that it saves labor; however, it is also used to save energy and materials and to improve quality, accuracy and precision.

6. **Access Control:** In the fields of physical security and information security, access control is the selective restriction of access to a place or other resource. The act of accessing may mean consuming, entering, or using. Permission to access a resource is called authorization. Locks and login credentials are two analogous mechanisms of access control.

7. **Auto theft:** Motor vehicle theft is the criminal act of stealing or attempting to steal a car (or any other motor vehicle).